An ERRWT Model for Slang Prediction

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March 2026

Abstract

We use the Edge Reinforced Random Walk with Triggering (ERRWT) model proposed in [3] to identify slang words and study quantiative linguistic laws in a twitter dataset. ERRWT is constructed from the random walk model and includes edge-reinforcement as edges are traversed and introduces the concept of triggering nodes. Zipf's Law, which describes the power law dynamic of a rank frequency diagram of a corpus of text, and Heaps' law, which studies the relationship between the number of words and the number of distinct words in a corpus of text, are evaluated. We find ERRWT is largely ineffective at identifying slang words across the dataset, but this is partially due to limitations on the effectiveness of the dataset to be a representative body of the English language. We find the data can be shown to be quasi-Zipfian and does observe Heaps' law. Future work should examine larger datasets over an extended time period to better evaluate the temporal element of the model, and additional constraints should be added on slang identification to improve hit rate. Overall, this project expanded on the Edge-Reinforced Random Walk with Triggering model proposed in [3]. This model is constructed on the original basic model of the simple random walk, introduces the concept of reinforced edges with probabilities that improve as the edge is traversed more frequently, and lastly adds a triggering concept that allows for the discovery of new nodes and the addition of surrounding reinforced edges. We studied a dataset of 1.6 million tweets sourced over a year, and tested the triggering component by partitioning tweets into two groups to construct a base set and allow for triggered nodes in the second set. We compared our analysis against a base case model that used an English language dictionary to identify slang words versus English words. When studying both models, both outputs obeyed a quasi-Zipfian distribution and obeyed Heaps' law, but failed to effectively and consistently identify slang words. Future work should expand on data sources and parameter tuning to create a fair balance of base words and triggered words, and the temporal range of the data should be extended to effectively look for trends over time using the model.

1 Background

1.1 Overview of Random Walks

The foundation of the Edge Reinforced Random Walk rests in the Simple Random Walk. Suppose a person steps out of the bar with the intention of returning home. Between their house and the bar is n streets, so a direct path home will take n "steps" meaning streets traversed. Also suppose these n streets are arranged exactly in a line, with the house at one end and the bar at the other end. Due to inebriation, this person decides to flip a coin at every intersection to decide where to go next instead of following the direct path home. Head's will have them go in the direction of the house, while tails will have them walk in the direction of the bar, at least until they run out of streets. Therefore they have $\frac{1}{2}$ probability of walking right and $\frac{1}{2}$ probability of walking left. How many streets will the person need to walk in order to get home?

This doesn't have a simple answer. In fact, with no restriction on the number of steps being taken, there are infinitely many paths that the person may take. In fact, if the person were to return home in the minimum number of steps, they have a $\frac{1}{2^{n-1}}$ probability of ending up at their house in that time. However, due to the constraints of this one street town, we know the person, will eventually make it home. In fact, even with a weighted coin having unequal probabilities of heads or tails, as long as the probability of heads is nonzero, they will certainly return home. This remains true if the town were arranged on a grid due to the unlimited number of steps that can be taken.

Random walks are prevalent in modeling biological systems, particle diffusion, and stock price fluctuations. However, we know that many systems don't behave with this simple random behavior. In fact, there are often hidden drivers that directs a random walk towards some point, like a stock approaching its equilibrium price. Due to this, some random walk models have weights or reinforcing factors that shift the probability. Suppose that once the person stumbles halfway home from the bar, they recognize the space and have a higher likelihood of returning home. In cases of biological systems, multiple walkers may take random but eventually similar paths, creating a deterministic system from a random one. Hence, the edge-reinforced random walk was formulated in [2]

The underlying model for this research is the Edge-Reinforced Random Walk with Triggering (ERRWT), first introduced in [3]. This introduces the concept of novelty, where new nodes are discovered over time. Using this idea, we may expand the capabilities of our modeling into predicting slang and studying other linguistic trends.

1.2 Quantitative Linguistics

Quantitative linquistics (or statistical linguistics) is focused on language learning, language change, and and structure of natural languages. Using statistics, we may investigate language structures and determine trends. Quantita-

tive linguistics also leads to a slew of different empirical laws that demonstrate the mathematical relationships. These include Zipf's law, which describes the ranked frequency of words as an inverse power law relationship [1], and Heaps' law, which describes the relationship between the number of distinct words in a text with the length of the text [5]. Heaps' law is shown to be asymptotically identical to Zipf's law [6], but can be distinctly evaluated, with Heaps' law being shown in English languages within a certain band of coefficients of the regression, and Zipf's law being shown through finding a power law distribution. We can evaluate how these laws are observed in our dataset, and critically evaluate if these laws should hold true for something similar but distinct from an English corpus.

2 Approach

The objective of this research is to study how the ERRWT is capable of identifying novel slang words used in current English language. To do this, we will compare our model's results against the results produced by a base case that identifies words as English or non-English. In addition, we want to identify how our data behaves in response to different linguistic laws. Part of this means comparing behavior of words identifiable in the dictionary as well as behavior of the slang words we identify. This will guide us in answering questions about slang based in English and if it can be perceived on the same pedigree as classic English.

2.1 Dataset Source

The data studied in this project is the Sentiment140 dataset, publicly accessible on Kaggle [4]. It contains 1.6 million instances of twitter data sourced using the now defunct twitter API. All tweets were published in 2009, and the data is chronological by date. The intention of the dataset is sentiment analysis used in ML and AI training projects, but we will ignore the sentiment and focus entirely on the linguistic implications of the tweets.

2.2 Objectives of Linguistic Study

It's worth noting that tweets are limited to only 280 characters in length, so any dataset of tweets will have some upper bound for the number of words. In addition, since a tweet only consists of at most a few sentences, finding instances of repeat words may be much rarer than in longer bodies of text, such as those found in classic literature. Also, this limit may lead users to shorten or contract many sentences, or intentionally misspell words. This restricts our ability to analyze tweets, and the dynamic components of language prevent us from complete analysis. Due to this, we need to use natural language processing (NLP) software to make our analysis interesting.

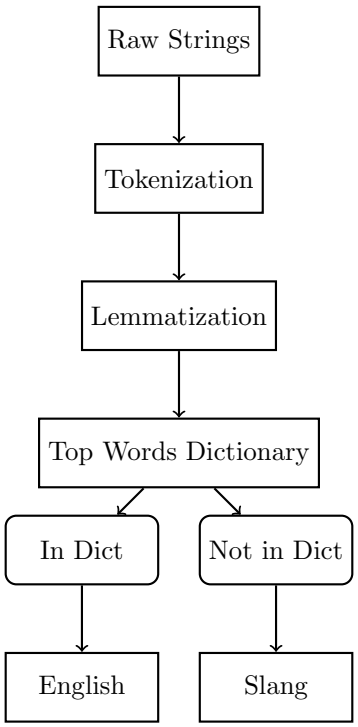


Figure 1: Base Model Flowchart

2.3 Natural Language Processing

Quantitative linguists often use lemmatization to simplify words into their base form. This involves trimming endings, reducing variations, and reversing tenses.

$$\text{running, ran, runs} \rightarrow \text{run}$$

We may achieve this using a lemmatization function from the Python NLTK (Natural Language Toolkit) library, which takes a word and compares its subsets against WordNet, a dictionary of known words, and maps words to their appropriate lemmas. The only issue is WordNet has no ability to handle misspellings, so some words aren't correctly mapped or aren't mapped at all.

$$\text{runing} \not\mapsto \text{run}$$

Fortunately, lemmatization in python returns the original word if it doesn't map correctly instead of dropping the word.

Implementation will require us to take strings, split on spaces into tokens, and lemmatize each token using NLTK. There are a few key pitfalls with this approach. Due to the lack of mapping for non-English slang words, they will

be comparatively underrepresented in popularity to English words. However, these tools enhance our ability to properly identify linguistic trends. Identifying key "slang" tokens through this process will allow us to properly identify these words and guide us towards a better understanding of the English language.

3 The ERRWT Model

Next, we need to evaluate the models used in this study and create the link between the models and the data.

3.1 Base Comparison

As a base case comparison, we build a dictionary of the top 50000 words to evaluate our tokens, and only allow slang identification for words not identified in our dictionary. This creates a simple, cheap model that identifies slang words when they cannot be recognized, and should evaluate words on the fringe of popularity as slang. This flow is shown in Figure 1.

3.2 Edge-Reinforced Random Walks with Triggering

First, we need to evaluate the basic ERRW model from which we construct our used model.

Definition 1 (Edge-Reinforced Random Walk [2])**.** Let $(X_n)_{n\in\mathbb{N}}$ be a random process taking values in the set of vertices $V(G)$ on graph G, and let $F_n = \sigma(X_0, X_1, X_2, ..., X_n)$ be the filtration of its past steps. For any edge $e \in E(G)$ and $n \in \mathbb{N}$, let

$$Z_n(e) = \sum_{k=1}^{n} \mathbb{1}_{\{X_{k-1},X_k\}=e} + a_e$$

be the number of crosses of e up to n steps with initial weight ρ_e Then $(X_n)_{n\in\mathbb{N}}$ is an Edge Reinforced Random Walk (ERRW) with starting point $v_0 \in V(G)$ and weights $(\rho_e)_{e\in E(G)}$. If $X_0 = v_0$, then for all $n \in \mathbb{N}$:

$$\mathbb{P}(X_{n+1} = j|F_n) = \mathbb{1}_{j\sim X_n} \frac{Z_n(\{X-n, j\})}{\sum_{k\sim X_n} Z_n(\{X_n, k\})}$$

This model takes our initial perspective on the randomness in random walks and makes it dynamic, because additional points are added to reward used edges. The simple random walk doesn't encapsulate this, because edge probabilities remain constant and don't shift over time. With this, edge probabilities become their own random variables, creating a new realm of complexity that is able to address many real world modeling constructions in physics, finance, and biology. To apply this model to the complex and time-sensitive nature of language, we must enable the triggering of new nodes.

Definition 2 (Edge-Reinforced Random Walk with Triggering [3])**.** Suppose at time t, the graph $G^t = (V^t, E^t)$ with weighted adjacency matrix $W^t = \{w_{ij}^t\}$. The Edge Reinforced Random Walk $(X_{m,n})_{m,n\in\mathbb{N}}$ has probability:

$$\mathbb{P}(X_{m,n+1} = j | X_{m,n} = i) = \frac{w_{ij}^t}{\sum_{\{ik\}\in E^t} w_{ik}^t}$$

and reinforcement:

$$w_{ij}^{t+1} = w_{ij}^t + 1$$

if the edge is chosen. If an edge is triggered between two nodes on step n' of random walk $(X_{m,n})_{m\in\mathbb{N},n\leq n'}$, we have the following graph:

$$G^{t+1} = (V^{t+1}, E^{t+1}) = (V^t, E^t \cup \bigcup_{1\leq p\leq n'} \{pl\})$$

with weights:

$$w_{pl}^{t+1} = 1, p \leq n'$$

If a new node l is reached on step n' of random walk $(X_{m,n})_{m\in\mathbb{N},n\leq n'}$, we have the following graph:

$$G^{t+1} = (V^{t+1}, E^{t+1}) = (V^t \cup \{l\}, E^t \cup \bigcup_{1\leq p\leq n'} \{pl\})$$

with weights:

$$w_{pl}^{t+1} = 1, p \leq n$$

With these additions, we have the Edge Reinforced Random Walk with Triggering $(X_{m,n})_{m,n\in\mathbb{N}}$ on graph G^t.

Now, we may articulate the data flow of our project. Similar to the base case, we will be tokenizing strings and lemmatizing them. However, we then construct a basis dictionary using the first 20% of the dataset. This is the jumping off point for our graph, and will make up the words that we identify as English. After, we will work through the remaining dataset to identify any words with novelty, or words that aren't part of the initial dataset. This functions as the random walks that will trigger new nodes and identify new elements of the language. The flowchart is shown in Figure 2.

4 Results

Now we may analyze how our two models performed on the data.

4.1 Zipf's Law

To evaluate the ability of our data to fit the power law distribution, we use the method proposed in [1]. This involves fitting the parameters $x_{\min}$ and α.

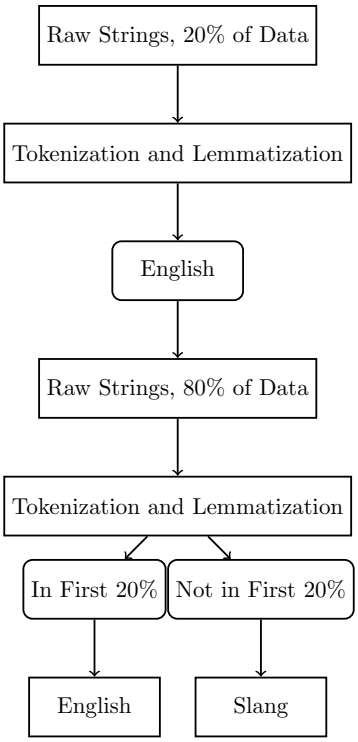


Figure 2: ERRWT Model Flowchart

We find $x_{\min}$ by minimizing the Kolmogorov-Smirnov statistic for a given CDF $D_n = \sup_x |F_n(x) - F(x)|$. We then estimate α using the maximum likelihood estimation (MLE) formula:

$$\hat{\alpha} = 1 + \frac{n}{\sum_{i=1}^{n} \ln \frac{x_i}{x_{\min}}}$$

Finally, we compute the Kolmogorov-Smirnov statistic D using the maximum difference between the empirical CDF of our data $S(x)$ and the hypothetical power law CDF $P(x)$. Lastly, we simulate power law datasets and repeat the same process, calculating a D value for each. The p-value is the fraction of simulated values that are larger than our observed Kolmogorov-Smirnov statistic. For each dataset, we used 2500 simulations and used the powerlaw Python library to compute automatically. Key statistics are shown in 1.

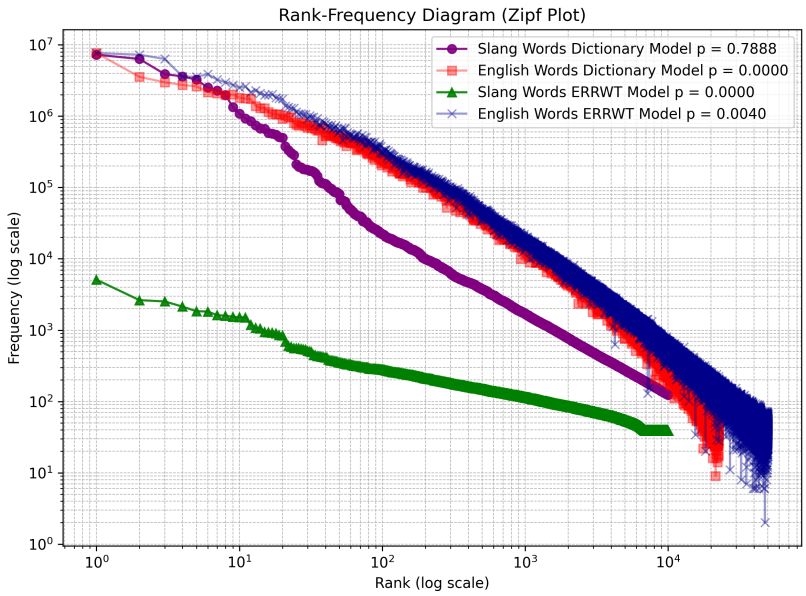


Figure 3: Zipf Plot for Both Models Partitioned by English and Slang

Dataset	α	$x_{\min}$	D
Slang Dictionary	1.8769	125.0000	0.0058
English Dictionary	1.7775	8763.0000	0.0238
Slang ERRWT	2.9724	49.0000	0.0781
English ERRWT	1.6082	52.0000	0.0080

Table 1: Computed Values for Kolmogorov-Smirnov Test, $N = 2500$ Simulations

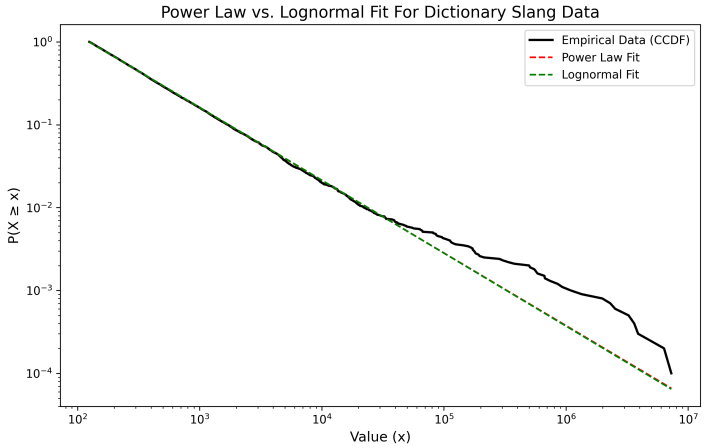


Figure 4: CCDF of Slang English Dictionary Model Data

As shown in Figure 3, we produced low p-values for every dataset except for the English Dictionary slang dataset. For this dataset, we must take one last step by comparing its fit to a lognormal distribution using a log-likelihood test. We found the test to be inconclusive and neither fit to be significantly better ($R = 0.9979, p = 0.3183$.) The Complementary Cumulative Distribution Function of this is shown in Figure 4.

4.2 Heaps' Law



(a) Heaps' Law Under ERRWT (b) Heaps' Law Under Dictionary

Figure 5: Two Different Observations of Heaps' Law. Each instance is treated as a corpus of text. Both graphs are identical except for slight differences in slang identification as shown in color. Both models produced an average slang identification rate of 3%.

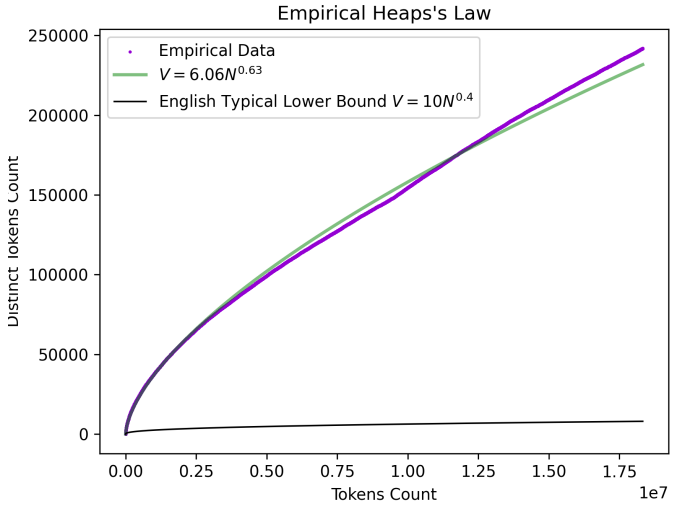


Figure 6: Heaps' Law Where Tweets are Gathered Contiguously and Treated as One Body of Text. The typical lower bound for English texts is $K = 10, \beta = 0.4$ and the upper bound is $K = 100, \beta = 0.6$ [5].

Here, we discuss two different results. In Figure 5 we treat each individual tweet as its own body of text, and attempt to observe Heaps' Law in that behavior. In addition, the gradient shows the fraction of the tweet that is identified as slang.

In Figure 6, we instead treat the entire dataset as one body of text and fitted the exponential regression $V = 6.06N^{0.63}$. For a given Heaps' law fit $V = KN^{\beta}$, the typical lower bound observed in English language texts is $K = 10, \beta = 0.4$ and the upper bound is $K = 100, \beta = 0.6$ [5].

Both models produced an average slang identification rate of 3%. Neither model produced a high hit rate for true slang words, and a lot of noise was produced in both models. For the Dictionary model, lowercase forms of words like "friday" and "sunday" were identified as slang, as well as many articles and conjunctions such as "and", "but" and "to". In the ERRWT model, many usernames were identified as slang, as well as a wide range of various parts of speech.

5 Discussion

Recall that Zipf's Law describes the rank and frequency of words across a language as inversely proportional. We describe this mathematically as an inverse power law relationship. However, we test whether our data follows this distribution by analyzing the log-log scale and looking for a linear relationship. We

also test this using the Kolmogorov-Smirnov test, and find no statistical significance for the data to necessarily fit the inverse power law distribution. Only the Slang Words from the Dictionary Model showed statistical significance, but we later observe in Figure 4 a deviation from the power law fit and the log-normal fit, meaning there is no motivating evidence to suggest this follows the Zipfian distribution. In fact, the concave down behavior in Figure 3 indicates a quasi-Zipfian distribution.

This can potentially be explained by the data partition, which leaves holes in the ranking and creates a steeper drop. However, Heaps' law is much more properly observed in Figure 6. While our estimated value for K is lower than the typical lower bound, the β is higher than the upper and the regression ultimately fits within the gap. This is to be expected, because the language used on Twitter doesn't precisely fit the language used in more traditional English texts. In fact, the structure of a tweet encourages brevity in both sentence length and word length, restricting the number of supposedly available words in one's vernacular. Common speech patterns are already predisposed to shorter phrases, simpler words, and acronyms. Tweet structure restricts this even further. While the dataset may be in spoken English, the traditional structure of an English corpus is morphed into a unique variety that may not obey the traditional structure.

Lastly, we found no major differences between the two models in the ability to identify slang correctly. Some overlap was found in what the two models studied, but they weren't necessarily very effective at distinguishing slang and normal english. The Dictionary Model struggled with various contractions and common parts of speech that didn't make their way into the dictionary, whereas the ERRWT model is more apt to identify some slang but is less deterministic and more dataset dependent. Both models made major errors in identifying twitter usernames and some common English words as slang.

Future work on this project should expand on a greater variety of linguistic laws to study. In addition, changing the input parameters and taking data across a longer period of time and from more varied sources will create a more diverse background on the accuracy of model implementation. A larger data project could also allow for deeper analysis at the hit rate of the model, such as comparing the suggested slang words against a dictionary of established slang words. In addition, the temporal element could be extended to a much longer time period. Sources like Google Analytics could be used to advance this project to focus on more particular instances of slang in certain linguistic subcultures or to study breakpoints in time that may have massively shifted linguistic trends. Regarding slang identification, increasing dataset size and using machine learning approaches, such as a random forest of random walk models, could be useful in building a more accurate mechanism with better stopgaps to correct errors.

6 Reflection

This project was essential in developing skills tangential to my major and to my development as a mathematician and as a scientific thinker. Getting to

critically examine language is such a fascinating venture that opened up many new research directions and advised me in what types of projects I enjoyed and wanted to focus on. Studying linguistics is highly engaging as a mathematician because it is deeply similar in its abstraction of familiar ideas. Taking a step back and questioning the very structure of language and the differences between languages offers a fascinating and engaging analytic framing of a deeply human idea. The opportunity to step over the STEM/humanities divide and explore language using mathematical ideas motivated me as a person and as a research. I felt that I was contributing to a new bridge in research that asks as many compelling questions as it conjectures.

Additionally, by introducing a temporal component to language and developing the mathematical ideas around it, I believe it introduced a sense of generational change and advancements in language that aren't identified in traditional linguistic models. I think this step in studying quantitative linguistics through the lens of change and non-constant behavior is essential to improving the quality of the study and the relevance of the field.

Quantitative linguistics is currently used in many AI and machine learning capacities to train chatbots and enable communication, and developing the ability to learn over time as human behavior grows and changes is essential for the improvement of these models. In addition, it will help advance mathematical models and the types of questions that are being asked in these frameworks.

7 Conclusion

Overall, this project expanded on the Edge-Reinforced Random Walk with Triggering model proposed in [3]. This model is constructed on the original basic model of the simple random walk, introduces the concept of reinforced edges with probabilities that improve as the edge is traversed more frequently, and lastly adds a triggering concept that allows for the discovery of new nodes and the addition of surrounding reinforced edges. We studied a dataset of 1.6 million tweets sourced over a year, and tested the triggering component by partitioning tweets into two groups to construct a base set and allow for triggered nodes in the second set. We compared our analysis against a base case model that used an English language dictionary to identify slang words versus English words. When studying both models, both outputs obeyed a quasi-Zipfian distribution and obeyed Heaps' law, but failed to effectively and consistently identify slang words. Future work should expand on data sources and parameter tuning to create a fair balance of base words and triggered words, and the temporal range of the data should be extended to effectively look for trends over time using the model.

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